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NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/01

Paper 1 Multiple Choice **29 September 2014**

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **15** printed pages and **1** blank page.

[Turn over

1 Which of the following statements about the cell is true?

A Endocytosis is an active process because energy is required to move materials into the cell against their concentration gradients.

B In simple diffusion, materials enter the cell down their concentration gradients whereas facilitated diffusion can move materials against their concentration gradients.

C In osmosis, water, ions and small polar molecules enter the cell by diffusing through a selectively permeable membrane down their concentration gradients.

D Materials such as ions and small polar molecules can enter the cell against their concentration gradient by active transport.

2 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

A Presence or absence of 180^o rotation of alternate subunits.

B Presence or absence of branching in the macromolecule

C Subunits are either monosaccharides or amino acids.

D Different tendency of macromolecule to form hydrogen bonds with water.

3 The flowchart below shows part of an enzyme-catalysed sequence which involves 4 compounds (**W** – **Z**) and 3 enzymes (**E1** – **E3**).

$$\text{Compound } \mathbf{W} \xrightarrow{\mathbf{E1}} \text{Compound } \mathbf{X} \xrightarrow{\mathbf{E2}} \text{Compound } \mathbf{Y} \xrightarrow{\mathbf{E3}} \text{Compound } \mathbf{Z}$$

The addition of another compound **M** results in no change in the concentration of compound **W**, an accumulation of compound **X**, and a near absence of both compound **Y** and compound **Z**.

What does this information indicate about the mode of action of compound **M**, and which is a method by which the production of compound **Z** can be increased in the presence of compound **M**?

	Mode of action of compound M	Method to increase production of compound Z
A	It inhibits **E1**.	Increase concentration of compound **X** in the reaction mixture.
B	It activates **E1**.	Increase concentration of **E2** in the reaction mixture.
C	It inhibits **E2**.	Increase concentration of compound **Y** in the reaction mixture.
D	It inhibits **E3**.	Increase concentration of compound **Y** in the reaction mixture.

4 The amount of DNA in a mammalian cell in early prophase I of meiosis is **X**. What is the amount of DNA in the same cell at G1 of Interphase ?

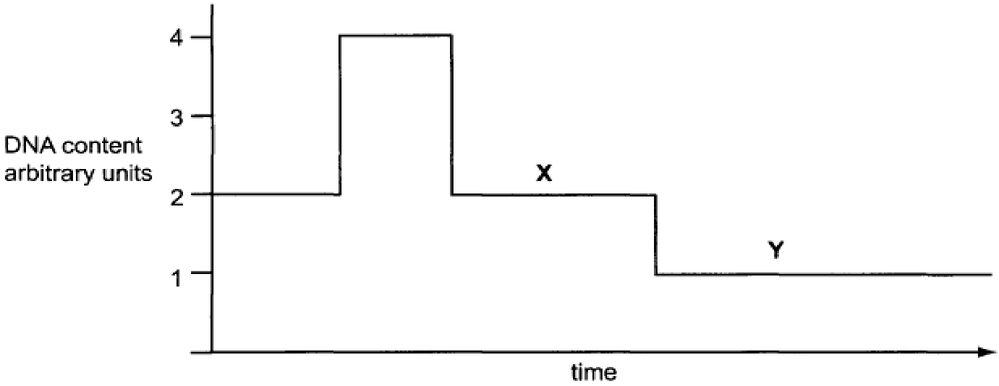
A 0.25**X**

B 0.5**X**

C **X**

D **2X**

5 The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



	Ploidy of cell at end of X	Ploidy of cell at end of Y
A	diploid	diploid
B	diploid	haploid
C	haploid	haploid
D	haploid	diploid

6 One individual, organism **P**, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

Normal protein	Variant protein
-gly-lys-lys-gly-glu-	**-gly-lys-gly-gly-glu-**

What could account for the presence of the normal protein and the variant protein in P?

1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.

2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.

3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.

4 The sequence of bases was altered in the DNA in a cell in an ovary of P.

5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.

A 1, 2 and 4 only

B 1 and 3 only

C 2, 4 and 5 only

D 3 and 5 only

7 Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.

Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?

A It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein.

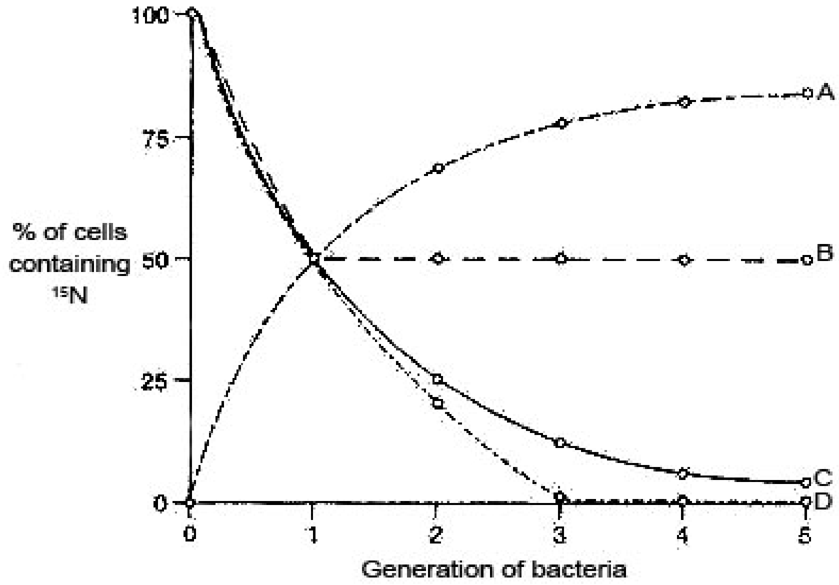
B It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein.

C It prevents translation by binding to the mRNA that contains information for the disease-causing protein.

D It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein.

8 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi conservative?



9 A polypeptide has the amino acid sequence glycine – arginine – lysine – serine.

The table below gives possible tRNA anticodons for each amino acid.

amino acid	tRNA anticodons
arginine	UCC GCG
glycine	CCA CCU
lysine	UUC UUU
serine	AGG UCG

Which sequence of bases on DNA would code for the polypeptide?

A CCACGCAAGAGC

B CCTTCCTTCTCG

C GGAAGGAAAAGC

D GGTTGGTTGTGC

10 The hepatitis C virus is a single-stranded, enveloped, positive sense RNA virus, which attack the liver cells of human beings. Which of the following general strategy states how the viruses enter the host cell?

A Uncoating of the viral envelope with the host cell membrane

B Injection of the whole virus into the host cell

C Fusion of the viral envelope with the host cell membrane

D Penetration of the viral capsid via budding

11 RNA viruses require their own supply of certain enzymes because

A host cells do not have enzymes available that can replicate the viral genome.

B the enzymes translate viral mRNA into proteins.

C the viruses use these enzymes to penetrate host cell membranes.

D these enzymes cannot be made in host cell.

12 The H1N1 virus is a subtype of the influenza A virus. Which statement about the H1N1 virus is true?

A The virus contains the enzyme reverse transcriptase.

B The virus comprises of a nucleocapsid enclosed by a membrane.

C The virus is not encased by a lipoprotein envelope.

D The virus genome is single-stranded DNA.

13 Which of the following characteristics or processes is common to both bacteria and viruses?

A Binary fission

B Nucleic acid as the genetic material

C Mitosis

D Ribosomes in the cytoplasm

14 A mutated *LacI* gene resulting in the failure of allolactose binding to the protein would cause

A reduced expression of operon.

B constitutive expression of the operon.

C reversible binding of protein to operon when lactose is present.

D permanent binding of protein to operon when lactose is present.

15 What are the correct characteristics for a prokaryotic genome?

	Promoters	DNA always bound to histone proteins	Plasmids often present	Repeat sequences absent or uncommon
A	✓	✗	✗	✗
B	✗	✓	✓	✓
C	✗	✓	✗	✗
D	✓	✗	✓	✓

16 Which feature is common in eukaryotic and prokaryotic genomes?

A Both genomes contain non-coding sequences.

B Both require general transcription factors to initiate transcription.

C The DNA in both genomes can be methylated to prevent hydrolysis by their own restriction enzymes.

D Both genomes are replicated during interphase just before binary fission or mitosis.

17 What is an example of translational control of gene expression?

A Activation of proteins by folding or enzymatic cleavage.

B Activation of chemical groups, such as phosphate groups, to free amino acids in the cytoplasm.

C Binding of protein factors to specific sequences in mRNA preventing ribosomes from attaching.

D Formation of disulfide bridges in the protein being formed.

18 What describes common features of the telomere and the centromere?

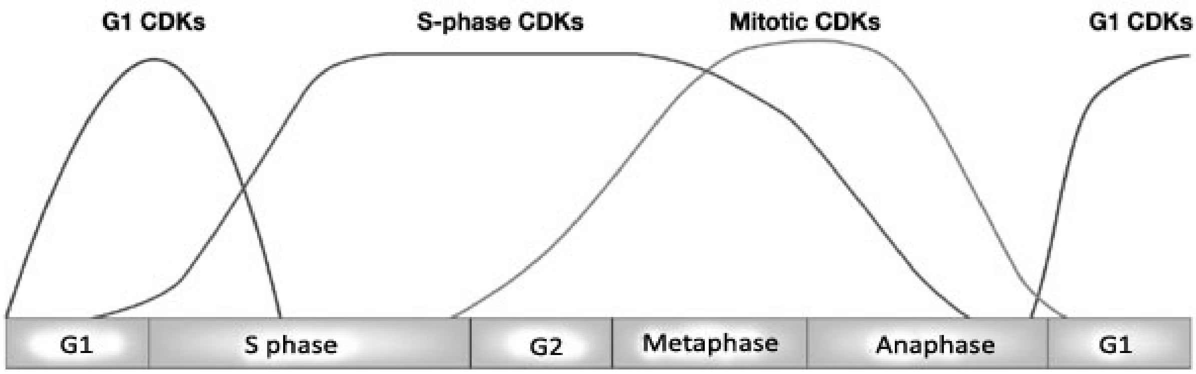
A Both areas have proteins associated with them.

B Both are made up of DNA with a high incidence of adenine and thymine.

C Both DNA sequences are conserved throughout the life of the organism.

D During interphase, DNA replication takes place simultaneously in both regions.

19 Cyclins are regulatory proteins that associate with cyclin-dependent kinases (CDKs) to control the different stages of the cell cycle. The right type and the correct amount of cyclins and CDKs must be present at the different stages to ensure regulation of the cell cycle. The concentrations of the different CDKs are shown in the diagram.



How could the levels of the different CDKs be regulated?

I Formation of heterochromatin.

II Binding of repressor to operator.

III Length of mRNA poly(A) tail.

IV Ubiquitylation of CDKs.

A **I** and **IV.**

B **II** and **III**.

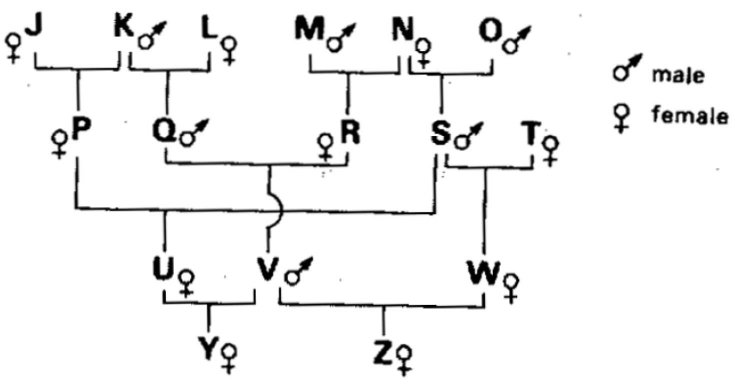
C **III** and **IV**.

D **I** and **II.**

20 The figure below shows a pedigree for the inheritance of the disease 'dropsy' in cattle. In this pedigree, both the calves Y and Z show symptoms of this disease. From previous experimental work, it is known that 'dropsy' is an inherited condition, due to a single pair of alleles, and is not affected by the sex of the animal.

Apart from the calves Y and Z, no other animals in the pedigree show symptoms of 'dropsy'.

You may assume that all animals are unrelated unless a relationship is shown in the pedigree.



Which one of the following great-grandparents carried a gene for 'dropsy'?

A N

B K

C M

D L

21 Fruit flies (*Drosophilia*), homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F1 and F2 generations were raised at three different temperatures. At each temperature, the F1 generation all had long wings.

The table shows the results in the F2 generation.

Temperature	***Result***
21ºC	¾ long wings, ¼ vestigial wings
26 ºC	¾ long wings, ¼ intermediate wing length
31 ºC	All long wings

Which statement explains these results?

A Wing length is under polygenic control.

B Long wing and vestigial wing illustrate codominance at 26 ºC.

C Long wing is dominant at higher temperatures but vestigial wing is dominant at lower temperatures.

D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

22 In garden peas, the allele T codes for terminal position of flowers and is dominant over the allele t, which codes for axil position.

In an experiment, 2 garden pea plants, both heterozygous for terminal flowers, produced 77 offspring with flowers at the terminal position and 43 offspring with flowers at the axil position. A student performed a χ^2 test to determine whether the results of the experiment follow the expected pattern of inheritance.

$$\chi^2 = \sum \frac{(O-E)^2}{E} \qquad df = n-1$$

df : degrees of freedom
n : number of classes
O : observed value
E : expected value

Distribution of χ^2

df	*Probability, p*				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following conclusions he made from the experiment and the χ^2 test is correct?

A Since $p < 0.05$, the flower position in garden peas does not involve a single pair of independently segregating alleles at the T locus.

B Since $p < 0.05$, the flower position in garden peas involves a single pair of independently segregating alleles at the T locus.

C Since $p > 0.05$, the flower position in garden peas does not involve a single pair of independently segregating alleles at the T locus.

D Since $p > 0.05$, the flower position in garden peas involves a single pair of independently segregating alleles at the T locus.

23 The table below shows the number of chromosomes in the parent species of the cabbage *Brassica oleracea*, the carrot *Raphanus sativus* as well as the various intermediate hybrids.

type of cell	***number of chromosomes per cell***
Parent cabbage	18
Parent carrot	18
Parent's gamete	9
F_1 hybrid	18
F_1 gamete	18
F_2 hybrid	36
F_2 gamete	18
F_3 hybrid	36

Chromosomal mutation takes place during the

A Fusion of parents' gametes

B Formation of F1 gametes

C Fusion of F1 gametes

D Formation of F2 gametes

24 In insects, sex is determined by a ZW chromosome system. Males are ZZ and females are ZW. A lethal recessive allele is sometimes present on the Z chromosome in moths.

What would be the sex ratio in the offspring of a cross between a male heterozygote and a normal female?

A 1 male : 1 female

B 2 male : 1 female

C 1 male : 2 female

D 3 male : 1 female

25 A metabolic poison rotenone was added to a mixture of homogenized cells with intact chloroplasts. It is discovered that rotenone can bind irreversibly to the electrons carriers on the thylakoid membrane in chloroplasts and alter their conformational shape.

In what way would the Calvin cycle be affected by rotenone poisoning?

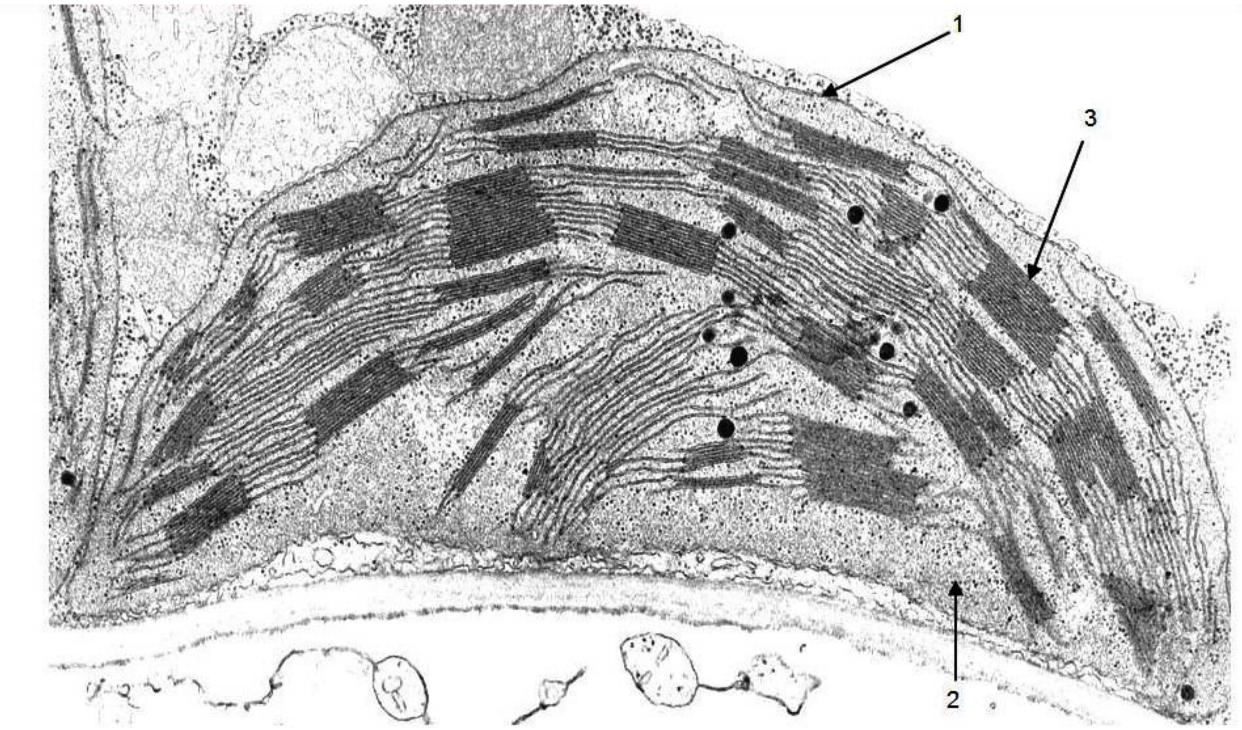
A Calvin cycle will stop, as the accumulation of triose phosphate will create feedback inhibition on glucose formation.

B The rate of Calvin cycle will decrease due to the accumulation of glycerate-3-phosphate.

C The rate of Calvin cycle will increase since electrons can flow directly to final electron acceptor to form reduced NADPH.

D There will be no effect since Calvin cycle is an enzyme-controlled process.

26 Which of the following is correctly associated with the function or characteristic of the labelled structure?



	Structure containing electron carrier	Structure where sugar is made	Structure with a double membrane
A	3	1	2
B	1	3	2
C	3	2	1
D	2	1	3

27 Certain drugs act at synapses and affect the action of neurotransmitter substances. The table shows the effects of four developmental drugs.

Drug	Effect
I	Fits snugly into the active site of cholinesterase
II	Induce the opening of voltage-gated Na^+ ion channels
III	Inhibits transfer of ions through the Na^+/K^+ pump
IV	Promotes the activity of proteins that allow Ca^{2+} ions into the pre-synaptic terminal

Which of the drug(s) would increase the chances of generating an inhibitory postsynaptic potential at the junction of an inhibitory synapse?

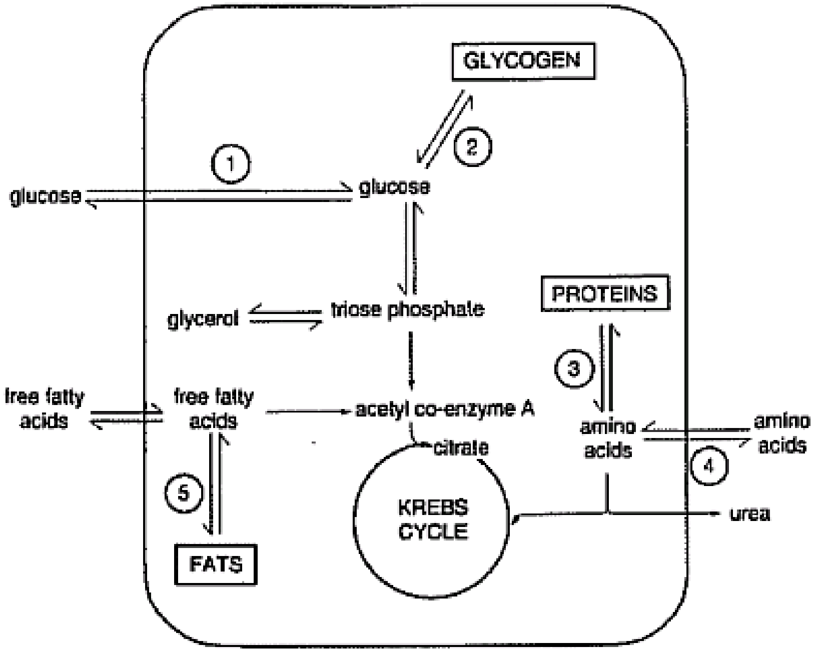
A I and IV only

B II and III only

C II and IV only

D I, II and III

28 The diagram shows some biochemical pathways in a liver cell. Some of the points where hormones affect the pathways are labelled 1 to 5.



At which numbered points would the hormone insulin accelerate the pathways in the directions indicated?

A 1, 2 and 3

B 1, 2 and 5

C 1, 3 and 4

D 2, 3 and 4

29 Which axon would transmit an action potential most rapidly?

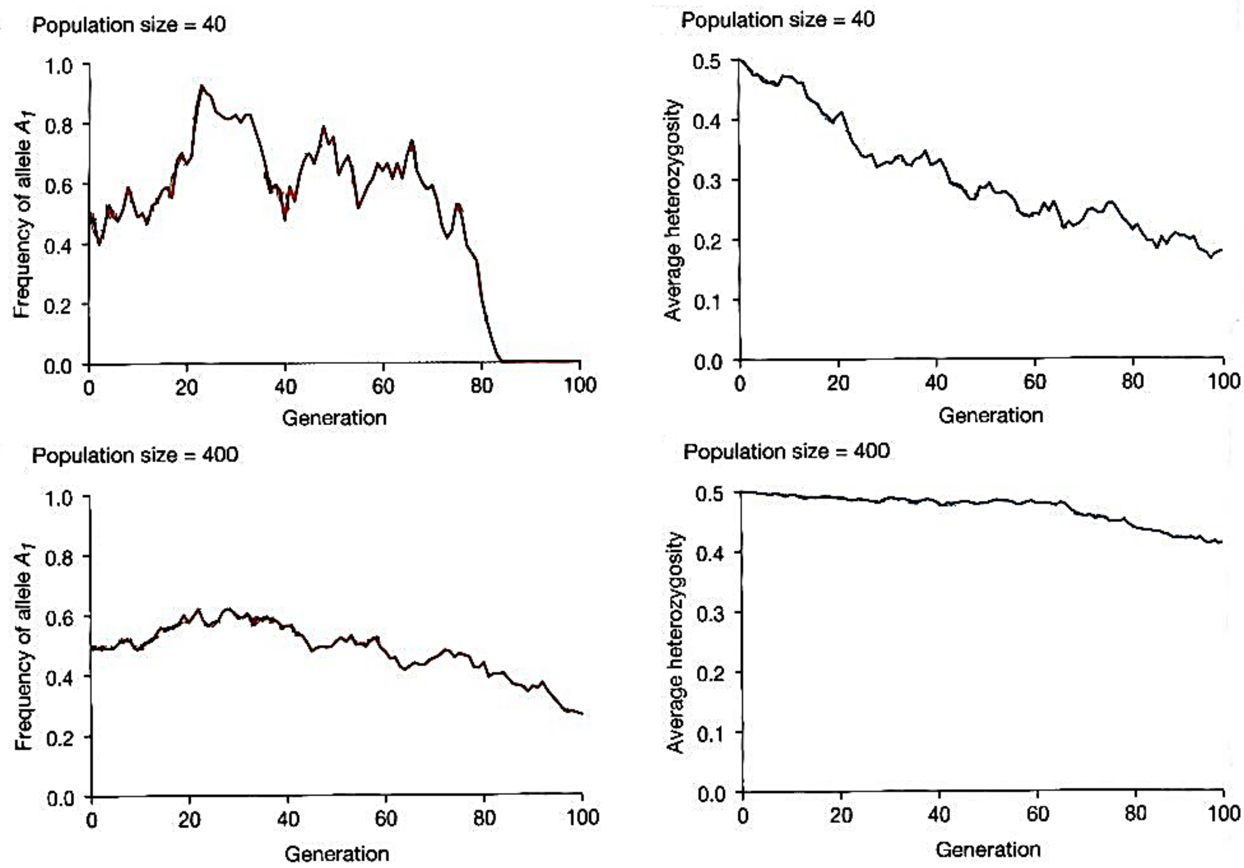
A 1 mm diameter neuron lacking myelin

B 1 mm diameter neuron with myelin

C 2 mm diameter neuron lacking myelin

D 2 mm diameter neuron with myelin

30 The four graphs represent simulations of genetic drift occurring in fruit fly populations of two different sizes, 40 and 400.



Which of the following about genetic drift can be deduced from the graphs?

A It leads to non-random fixation of alleles in the population.

B There is loss of heterozygosity in the population regardless of population size.

C It is a more powerful force of evolution in small populations.

D Given sufficient time, genetic drift can produce drastic changes in allele frequencies in large populations.

31 Which one of the following lists the main features of a phylogenetic classification system?

✓ represents presence of characteristic			
× represents absence of characteristic			
	common features	homologous structures	evolutionary relationships
A	✓	✓	×
B	✓	×	✓
C	×	✓	×
D	×	✓	✓

32 Modifications in the organization of the basic pentadactyl limb structure found in vertebrates provides good evidence for the principle of

A adaptive radiation

B convergent evolution

C genetic drift

D inheritance of acquired characteristics

33 An interbreeding population of finches became separated geographically, forming two isolated groups. Each group then became subject to different selective pressures. One group was then introduced into the habitat of the other.

Which one of the following would determine whether they now formed two distinct species?

A They had been separated for more than three million years.

B They failed to produced fertile F_1 hybrids.

C They showed marked differences in the shaped of their beaks.

D Several genes now possessed different base sequences.

34 The restriction enzyme *Dde*I recognizes a sequence on DNA, CTNAG (where N is any base). A portion of the β-hemoglobin sequence encoding normal hemoglobin is CCTGAGGAG.

If the mutation leading to sickle cell anemia had been CCTCAGGAG, would you be able to distinguish the two alleles with a restriction digest?

A Yes, the normal allele will be digested, the sickle allele will not.

B Yes, the sickle allele will be digested, the normal allele will not.

C No, both will be digested by *Dde*I.

D No, neither will be digested by *Dde*I.

35 A plasmid has two antibiotic resistance genes, one for ampicillin and one for tetracycline. It is treated with a restriction enzyme that cuts in the middle of the ampicillin gene. DNA fragments containing a human globin gene were cut with the same enzyme. The plasmids and fragments are mixed, treated with ligase, and used to transform bacterial cells. Clones that have taken up the recombinant DNA are the ones that

A are blue and can grow on plates with both antibiotics.

B can grow on plates with ampicillin but not with tetracycline.

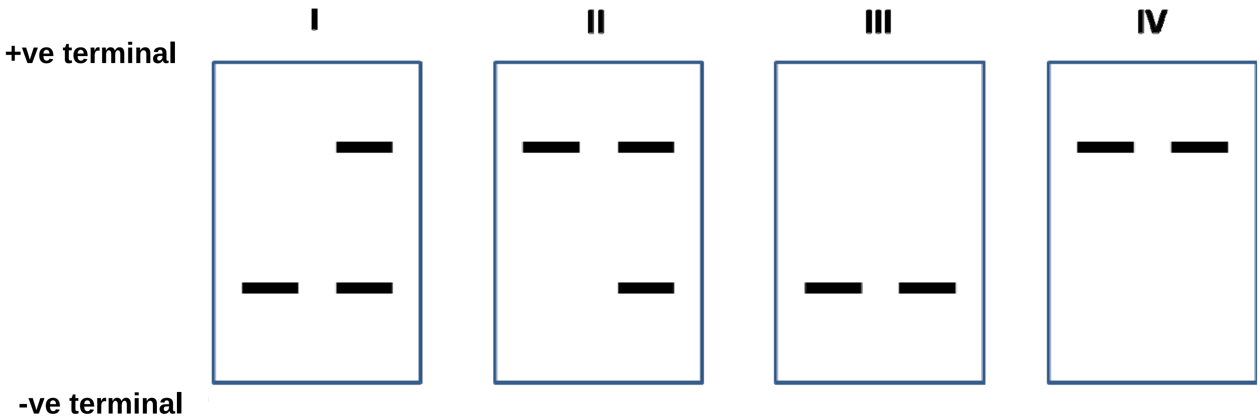
C can grow on plates with tetracycline but not with ampicillin.

D cannot grow with any antibiotics.

36 Cystic fibrosis (CF) is an autosomal recessive genetic disorder. An individual must have two copies of the mutated CFTR gene to express the disease phenotype. One of the most common CF-causing mutation resulted in a loss of phenylalanine located at position 508 of the protein.

The DNA sequence of the CF locus from the offspring of 2 carriers are removed and separated by gel electrophoresis.

Which pattern of bands corresponds to two of the offspring that are phenotypically normal?



A I only

B II only

C I and III

D II and IV

37 What is the key reason for using a greater range of probes and restriction enzymes in DNA fingerprinting?

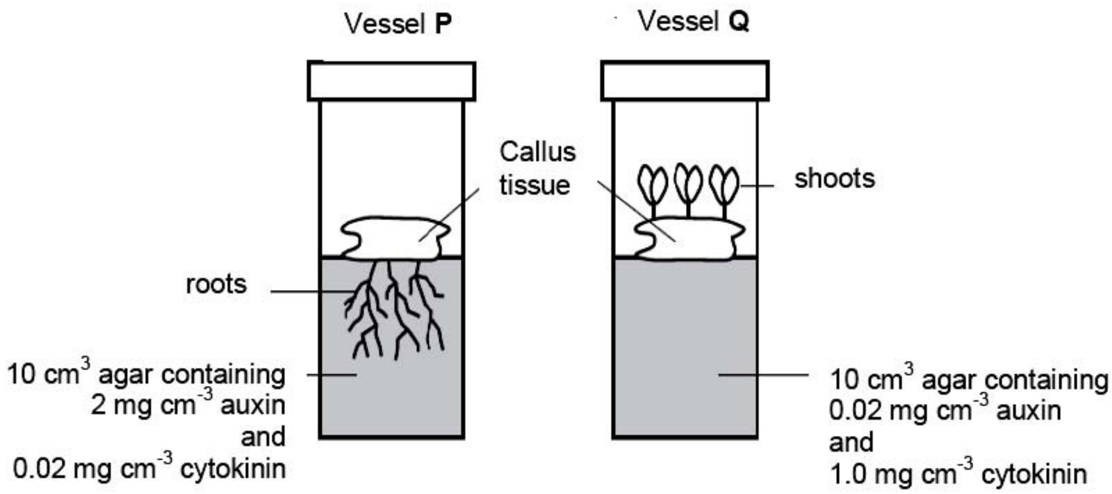
A It increases the likelihood that one of the probes will bind to the polymorphic region of the DNA after the latter is cut by the restriction enzymes.

B It is necessary for the creation of unique DNA fingerprints from two individuals' DNA that are significantly different in sequence.

C It permits more regions of the DNA to be analysed so as to reduce the possibility that two individuals' DNA would produce the same banding pattern.

D It allows all the DNA bands produced via restriction enzymes cutting to be detected so as to give a more accurate DNA fingerprint.

38 The figure shows the results of an experiment involving callus culture.



Which of the following statements is correct?

A A high auxin to cytokinin ratio promotes root instead of shoot formation.

B The cells making up the callus are uniform with no variations in morphology and physiology.

C The inner parts of the callus consist of dividing mass of newer tissue and in time will divide and become similar physiologically and genetically with the cells of the outer layer.

D The "knobby" appearance of the callus is because the cell division becomes unrestricted to all cells of the callus.

39 Which of the following is **not** a concern with growing genetically engineered *Bt* corn?

A Flow of the Bt gene from cultivated corn to wild relative thereby enhancing the weediness of certain corn relatives, which could then lead to loss of biodiversity.

B Transfer of marker genes that convey antibiotics resistance, used in the identification of successfully transformed cells, to bacteria that resides in the gut of humans thereby rendering the bacteria antibiotic resistant.

C Toxic effects of Bt on non-target lepidopteran insects (e.g. monarch butterfly larvae of the insect order *Lepidoptera*) could upset ecological balance.

D Transfer of Bt genes to the pests thereby increasing their resistance to the Bt toxin.

40 Which combination best describes a retroviral vector for use in gene therapy?

	expression	integrate into genome	transduction efficiency
A	stable	no	high
B	stable	yes	high
C	stable	yes	low
D	transient	no	low